

FRIDA

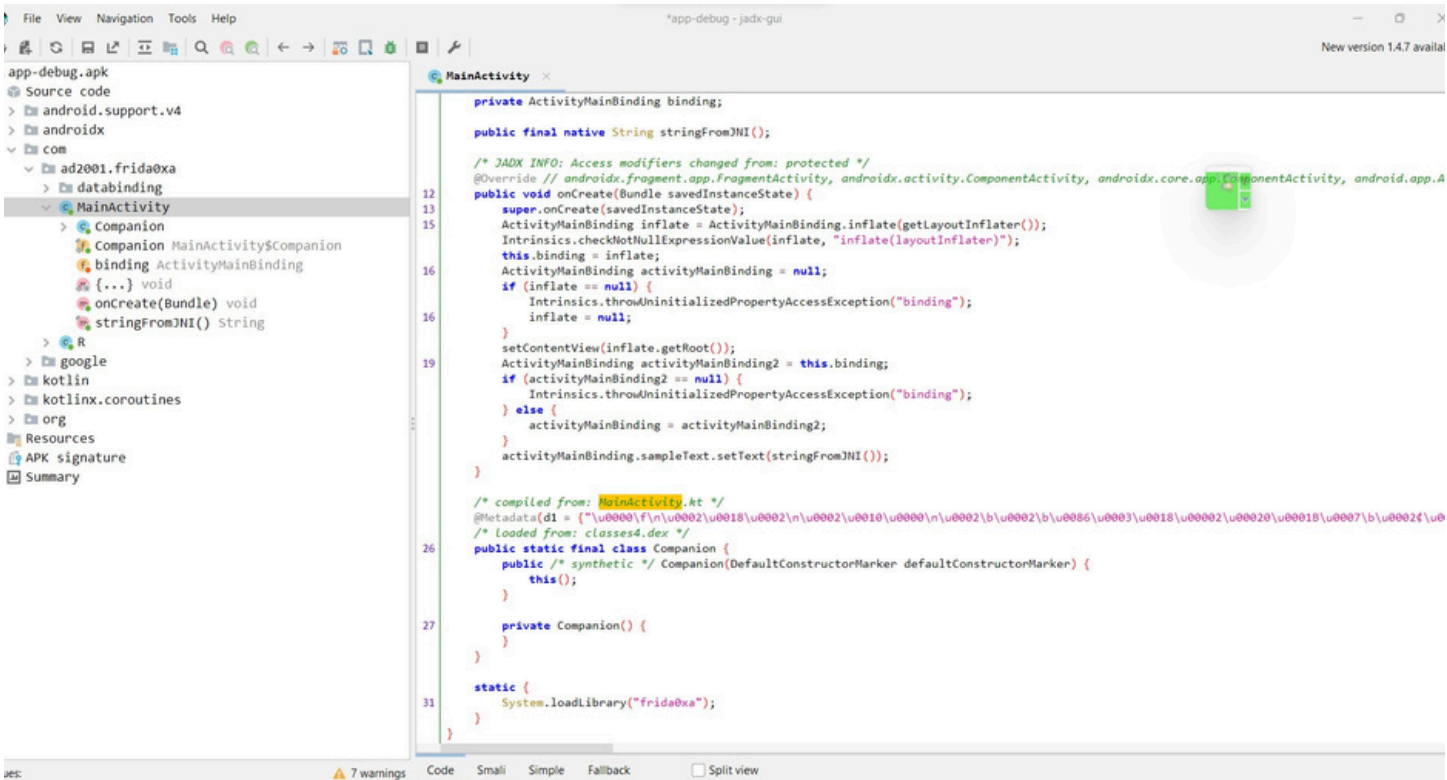
Frida LAB 10

Objective:

In Challenge 0x10, the application performs the validation inside a constructor. Instead of hooking a regular method, you must hook the class constructor and modify the object's state during initialization.

This challenge introduces hooking constructors using Frida.

Use jadx



This is exactly like our previous apk but this time the apk is developed in kotlin. We can see it's loading the `frida0xa` library at the bottom.

``j`ava

```
static {
```

```
System.loadLibrary("frida0xa");
```

}

///

We can also see a JNI method at the beginning.

```

11  @Metadata(d1 = {"\u0000&\n\u0002\u0018\u0002\n\u0002\u0018\u0002\n\u0002\u0018\u0002\n\u0002\u0002\b\u0002\n\u0002\u0002\u0018\u0002\u0002",
/* Loaded from: classes4.dex */
12  public final class MainActivity extends AppCompatActivity {
    public static final Companion Companion = new Companion(null);
    private ActivityMainBinding binding;

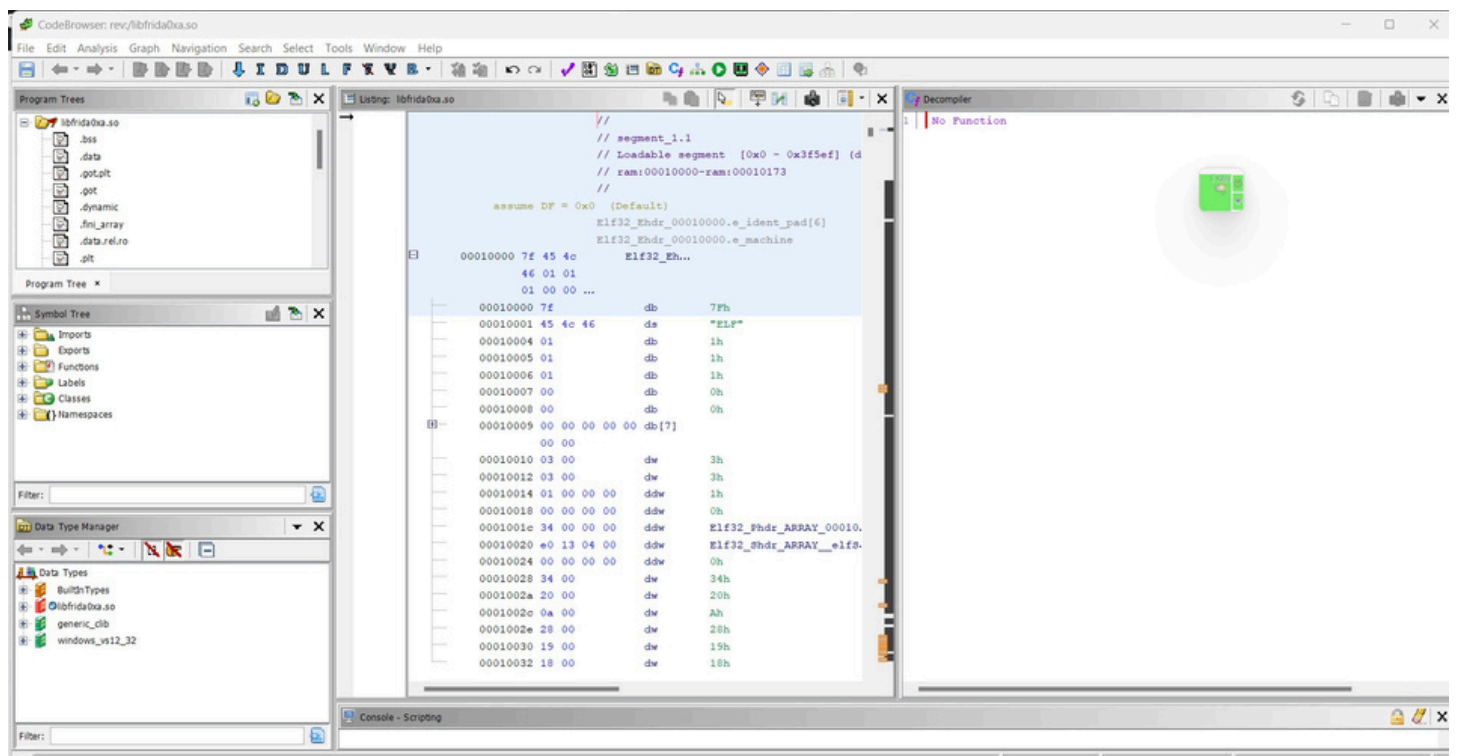
    public final native String stringFromJNI();

    /* JADX INFO: Access modifiers changed from: protected */
    @Override // androidx.fragment.app.FragmentActivity, androidx.activity.ComponentActivity, andro
13  public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

```

Decompile this apk using apktool .

Use ghidra.

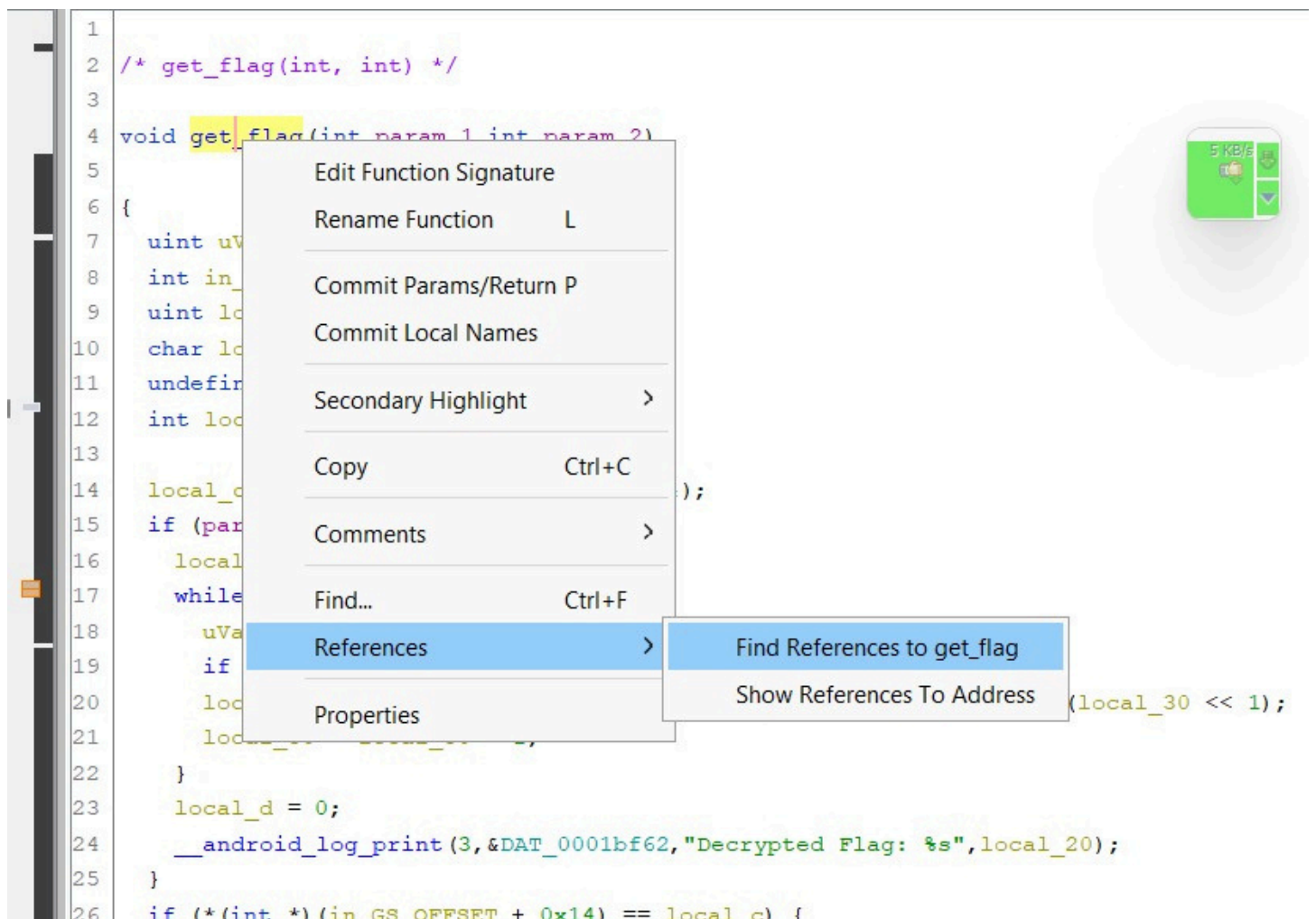


We have two functions.

- `get\_flag()`
- `Java\_com\_ad2001\_frida0xa\_MainActivity\_stringFromJNI()`

The next function is the `get\_flag()` method. This function wasn't declared in the Java space, and it's also not being called from anywhere in the library. You can confirm this by checking its cross-reference. The only reference is in the (Frame Description Entry) FDE table.

Let's examine the decompilation of this method.



frida -U -f com.ad2001.frida0xa

```
frida -U -f com.ad2001.frida0xa

16.1.5 - A world-class dynamic instrumentation toolkit

ds:
lp      -> Displays the help system
ject?   -> Display information about 'object'
it/quit -> Exit

nfo at https://frida.re/docs/home/

ted to Android Emulator 5554 (id=emulator-5554)
.frida0xa`. Resuming main thread!
554::com.ad2001.frida0xa ]->
```

Final output:

